FinSight PK

Regime-Aware, Sentiment-Augmented KSE-100 Forecasting
with Explainability & Web Deployment

Type	Hybrid — Development + Research
Domain	Financial AI / NLP / Time-Series Forecasting / Explainable AI
Team	3 Members | Semester 6 & 7 | FAST-NUCES Lahore
Target	IEEE INMIC / ICIET Pakistan (Semester 7) + Web Application

1. Problem Statement

Pakistani stock market participants — retail investors, portfolio managers, and fintech firms — make decisions on the KSE-100 with little to no AI-assisted tooling. Globally, hybrid forecasting systems combining news sentiment, macroeconomic signals, and deep learning are actively researched, yet no such system exists for Pakistan's equity market. Two critical gaps compound this problem:

- There is no labeled Pakistani financial news sentiment dataset.
- KSE-100 forecasting models ignore market regime context entirely — predicting bull and bear markets with the same model and features, which is fundamentally flawed.

FinSight PK addresses both gaps through an end-to-end AI pipeline combining NLP, regime detection, hybrid deep learning, explainability, and a publicly accessible web application.

2. Literature Review

The table below summarises 10 recent papers (2021–2025) directly relevant to FinSight PK's four pillars: financial NLP, hybrid LSTM forecasting, market regime detection, and explainable AI.

Reference	Problem Addressed	Dataset Used	Method	Results	Limitations
Zeng & Jiang (2023)	Sentiment-augmented stock movement forecasting	Stock market news corpus (English)	FinBERT + LSTM	Outperforms BERT, standalone LSTM, ARIMA	English news only; no regime context or multi-horizon forecasting
Dhokane & Agarwal (2024)	LSTM with technical indicators for next-day price	NSE India (2020–2023), Yahoo Finance	LSTM + RSI, MACD, Bollinger Bands	Improved accuracy over price-only LSTM	No sentiment, macroeconomic signals, or regime detection

Reference	Problem Addressed	Dataset Used	Method	Results	Limitations
Muhammad et al. (2024)	Explainable deep learning for 5-class stock trend prediction	DAX30, FTSE100, Nikkei	Deep learning + SHAP + LIME	94.9% accuracy; top-10 feature importance via SHAP	No sentiment or regime context; not tested on emerging/Pakistani markets
Manikrao et al. (2025)	Interpretable LSTM with attention for stock price forecasting	Multi-stock (US markets), 25-run evaluation	LSTM + Attention + SHAP + LIME	Consistently outperforms XGBoost and plain LSTM	No regime-conditioned SHAP analysis; ignores bull vs. bear feature shifts
Ghani & Ghani (2024)	EPU indices as predictors of Pakistan stock market volatility	KSE-100 daily data + US, UK, China & Pakistan EPU indices	GARCH-MIDAS + combination forecasting	US EPU is strongest predictor of KSE-100 volatility	Traditional statistical model only; no deep learning or sentiment integration
Hameed et al. (2022)	KSE-100 stock price prediction with news sentiment (PSX banks)	PSX bank stocks 2011–2021 + NLTK news sentiment	Univariate LSTM + BiLSTM with self-attention	Sentiment improves prediction accuracy over price-only models	Lexicon-based sentiment (NLTK); no transformer; individual stocks only
Srijiranon et al. (2022)	Noise-filtered hybrid LSTM with sentiment for stock forecasting	Thai Stock Exchange (SET) — daily data	PCA + EMD + LSTM + Sentiment Scores	Lower RMSE vs. plain LSTM; decomposition aids accuracy	No regime detection; no explainability component
Raza & Akhtar (2024)	Stock price prediction using ML and technical indicators on KSE-100	KSE-100, Yahoo Finance (Jan 2010 – Oct 2023)	ANN, SVM, Random Forest, LSTM + 27 technical indicators	ANN/SVM: 85% accuracy, RF: 84%, LSTM: 78%	No sentiment analysis, regime detection, or macroeconomic features

Reference	Problem Addressed	Dataset Used	Method	Results	Limitations
Kasture & Shirsath (2024)	Hybrid RNN-LSTM with sentiment for stock direction prediction	Multiple global stock indices	RNN-LSTM + financial news sentiment	Outperforms standalone LSTM and ARIMA	All market conditions treated identically; regime-adaptive forecasting acknowledged as open gap
Vallarino (2025)	LSTM + Transformer hybrid with FinBERT sentiment	Apple Inc. (AAPL) — 3 years of daily data	LSTM + Transformer Attention + FinBERT	$R^2 = 0.91$ on short-term forecasts	Single US large-cap stock; no macroeconomic features, regime conditioning, or explainability

Table 1: Summary of Related Work

3. Proposed Methodology

FinSight PK is an end-to-end AI pipeline that forecasts KSE-100 index movement using four sequential steps. Each step produces a concrete output that feeds directly into the next.

3.1. Step 1: Building a Pakistani Financial Sentiment Dataset & Training a Sentiment Model

The Problem. No labeled Pakistani financial news dataset exists. Existing systems use English or Indian data, which does not reflect Pakistan's market language or economic context.

What We Do.

- Scrape business news headlines from **Dawn Business** and **Profit.pk** — two of Pakistan's most widely read financial news sources.
- Manually annotate approximately **1,500 headlines** as Positive, Negative, or Neutral. For example:
  - *"KSE-100 surges 800 points on foreign inflows"* → Positive
  - *"State Bank raises interest rates to 22%"* → Negative
  - *"PSX trading volume remains flat"* → Neutral
- Fine-tune **DistilBERT** (a lightweight transformer model from HuggingFace) on this annotated dataset using an 80/10/10 train/validation/test split.

- Run inference on all historical headlines grouped by date.

What This Produces. A **daily sentiment score** for each trading day — representing whether that day's financial news was overall positive, negative, or neutral toward the market. Computed as:

$$sentiment\_score = positive\_ratio - negative\_ratio$$

This score becomes one input feature for the forecasting model in Step 3.

3.2. Step 2: Detecting Market Regimes from KSE-100 Price Data

The Problem. Most forecasting models treat all market conditions identically. Predicting prices during a bull market with the same model and features as a bear market is fundamentally flawed — the market behaves very differently under each condition.

What We Do.

- Download KSE-100 daily OHLCV data from **Yahoo Finance / PSX (2015–present)**.
- Compute two rolling statistics for each trading day:
  - **Rolling average return** (20-day window)
  - **Rolling volatility** — standard deviation of returns (20-day window)
- Apply **K-Means clustering** ($k = 3$) on the 2D space of (rolling return, rolling volatility) to assign every trading day to one of three regimes:
  - **Bull** — high return, low volatility
  - **Bear** — negative return, high volatility
  - **Sideways** — near-zero return, moderate volatility
- Validate cluster quality using the elbow method and silhouette score.

What This Produces. A **regime label** (Bull / Bear / Sideways) for every trading day. This label is one-hot encoded and passed as a feature to the LSTM in Step 3. This is the most novel component of the system — no prior KSE-100 paper integrates regime labels directly into the forecasting model.

3.3. Step 3: Forecasting KSE-100 Prices Using a Hybrid LSTM Model

The Problem. Standard price forecasting models only use historical prices. They ignore news sentiment, market regime context, and macroeconomic signals — all of which demonstrably affect Pakistan's equity market.

What We Do. Combine all previously computed features into a unified input matrix. Every row represents one trading day:

Feature	Source
KSE-100 close price (normalized)	PSX / Yahoo Finance
RSI (14-day)	Computed from OHLCV via `pandas-ta`
MACD	Computed from OHLCV
EMA	Computed from OHLCV
Daily sentiment score	Step 1 output
Market regime label (one-hot)	Step 2 output
Pakistan EPU Index	Publicly available dataset (monthly, forward-filled to daily)

This matrix is fed into an **LSTM (Long Short-Term Memory)** neural network — a model specifically designed for sequential time-series data — using a **30-day sliding window**: the model sees the last 30 days of all features to predict future prices. Training uses MSE/MAE loss, dropout regularization, and early stopping.

What This Produces. KSE-100 price forecasts at **7 days, 14 days, and 30 days** ahead, with prediction uncertainty estimated via Monte Carlo dropout.

3.4. Step 4: Explaining the Model's Predictions Using SHAP

The Problem. Deep learning models are often black boxes — they produce predictions but give no explanation for why. This is unacceptable for financial decision-making, where practitioners need to understand what is driving a forecast before acting on it.

What We Do. Apply **SHAP (SHapley Additive exPlanations)** — a mathematically grounded technique that assigns each input feature a contribution score for every individual prediction. We use `shap.DeepExplainer` on the trained LSTM model.

Critically, we run this analysis **separately per market regime**:

- *Bull regime predictions* → which features dominate when the market is rising?
- *Bear regime predictions* → does EPU or sentiment matter more during downturns?
- *Sideways regime predictions* → is RSI more influential when the market is flat?

What This Produces. Visual feature-importance plots per regime — bar plots and beeswarm plots showing which signals (news sentiment, EPU, RSI, regime label) are driving today's forecast and by how much. For example: *"In bear regimes, the EPU index has 3× higher SHAP magnitude than sentiment — suggesting macro uncertainty dominates over news tone during downturns."*

This per-regime SHAP analysis is what distinguishes FinSight PK from prior explainability work, which applies SHAP globally across all market conditions.

3.5. Dataset Summary

Data Source	Content	Usage
PSX / Yahoo Finance	KSE-100 OHLCV — daily (2015–present)	Core price time series
Dawn Business / Profit.pk	∼1,500 scraped headlines — manually annotated	Sentiment corpus & DistilBERT fine-tuning
Pakistan EPU Index	Monthly economic policy uncertainty index	Macroeconomic feature for LSTM
Computed Indicators	RSI, MACD, EMA — derived from OHLCV	Technical feature set for LSTM

3.6. End-to-End Data Flow

Dawn/Profit.pk Headlines → DistilBERT → Daily Sentiment Score ⟶ }
KSE-100 Prices → K-Means Clustering → Regime Label ⟶ } LSTM → Forecasts → SI
KSE-100 Prices → RSI / MACD / EMA ⟶ }
Pakistan EPU Index ⟶ }

4. How FinSight PK Differs from Prior Work

None of the surveyed papers simultaneously combine all four of the following in a single system for Pakistan's equity market:

- **Regime-conditioned forecasting.** No prior KSE-100 model integrates K-Means regime labels as a direct feature into an LSTM.
- **Pakistan-specific NLP dataset.** No labeled Pakistani financial news sentiment corpus exists. Prior work uses NLTK lexicons or English corpora.
- **Per-regime SHAP analysis.** Existing XAI studies apply SHAP globally. FinSight PK reveals which features dominate per market regime.
- **EPU + Sentiment + Regime unified.** No prior model combines Pakistan EPU, DistilBERT sentiment, and regime context in one LSTM pipeline.
- **Web application deployment.** FinSight PK delivers forecasts through a publicly accessible web interface, making AI tooling available to Pakistani retail investors and fintech practitioners.

5. Web Application

Beyond the research pipeline, FinSight PK will be deployed as a publicly accessible web application, making AI-assisted KSE-100 forecasting available to retail investors, portfolio managers, and fintech practitioners in Pakistan.

Feature	Description
Live KSE-100 Dashboard	Interactive chart displaying current index price, rolling volatility, and detected market regime (bull/bear/sideways) in real time.
Multi-Horizon Forecasts	7-day, 14-day, and 30-day KSE-100 price forecasts generated by the trained LSTM model, displayed with confidence intervals.
Sentiment Feed	Daily aggregated sentiment scores from Dawn Business and Profit.pk headlines, showing positive/negative/neutral breakdown.
SHAP Explainability Panel	Visual SHAP feature-importance plots showing which signals (sentiment, EPU, RSI, regime) are driving today's forecast — regime-conditioned.
Responsive UI	Mobile-friendly interface designed for Pakistani retail investors with plain-language forecast summaries alongside technical charts.
REST API	Lightweight API endpoints exposing forecast data and sentiment scores, enabling integration with third-party fintech applications.